

EDUCATION

University of North Carolina at Chapel Hill | Kenan-Flagler Business School – Chapel Hill, NC May 2029

Bachelor of Science, Business Administration; Bachelor of Science, Data Science | GPA: 4.0

- Honors: Assured Enrollment to Kenan-Flagler Business School; Honors Carolina: Top 10% of undergraduates

The Lawrenceville School – Lawrenceville, NJ

May 2025

High School Diploma | GPA: 3.92/4.0 UW

- Honors: Cum Laude Society; Herman Hollerith Prize Winner; Economics Club: NJ State Champion, National Finalist
- Publications: 2025 Journal of Future Economists (NY Federal Reserve): selected as top 12 of 195 papers nationwide

EXPERIENCE

Cofounders Capital – Cary, NC

January 2026 - Present

Venture Capital Intern

- Evaluated 100+ seed-stage B2B SaaS companies, analyzing market size, unit economics, competitive dynamics, and downside risk drivers to form investment views
- Synthesized diligence into reports assessing capital efficiency, risk-return tradeoffs, and exit viability for a \$60M fund

UNC Portfolio Management Team

September 2025 - Present

Technology Sector Analyst: 1 of 12 Students Selected from 170+ applicants

- Conduct TMT equity research for the Ewing-Schleeter Harris Fund (\$120K AUM, ~25% of portfolio), analyzing recurring revenue models, software transition dynamics, and margin expansion to develop buy/sell recommendations
- Build DCF models, WACC calculations, and public comparable analyses, stress-testing discount rates, growth assumptions, and exit multiples to triangulate price targets via football field valuation

Hitech Corporation – Gaithersburg, MD / Mumbai, India

July 2025 - September 2025

Marketing & Strategy Intern

- Designed U.S. market entry strategy via an automotive plastics acquisition, mapping supply chain tiers and identifying high-value OEM partners across 27 Tier-1 and aftermarket suppliers
- Analyzed competitor pricing and distribution models to design a 3-phase go-to-market strategy targeting \$50M+ TAM

DTV.ai – Boston, MA

May 2023 - February 2025

Co-founder of a B2B cost analytics firm

- Built and delivered 10 cost breakdown reports identifying sourcing inefficiencies, design redundancies, and 12–15% improvement opportunities in per-unit economics
- Managed client relationships and negotiated contracts to generate \$40K+ in revenue from retail and CPG partners

PROJECTS & RESEARCH

Working Paper (Quantitative Finance) | Research

March 2025 - Present

Title: American Option Pricing Under Time-Varying Rough Volatility | arXiv (arxiv.org/abs/2508.07151v2)

- Developed an American option pricing model integrating rough volatility dynamics and regime-switching engines to improve pricing accuracy; under review for journal publication
- Built fast signature-kernel approximations to reduce computational overhead, applying methods from stochastic calculus and machine learning to derivatives pricing

High-Frequency Trading Order Book | Independent Project

November 2025

Link to Repository: <https://github.com/roshanshah11/vertexladder>

- Developed an HFT order book in C++ processing 1.87M+ orders/sec, simulating the execution infrastructure used in institutional trading environments
- Optimized matching engine performance to minimize execution latency and slippage, applying financial markets engineering principles to market making workflows

VoiceBraille | M&TSI – University of Pennsylvania

July 2024

Portable Speech-to-Braille Printer as part of Penn's M&TSI Program (~5% acceptance rate)

- Led a 4-person team through concept generation, CAD prototyping (SolidWorks), and financial modeling to design a portable speech-to-Braille printer; earned "Most Innovative Idea" Award